DEVIXO SOLUTIONS

AI/ML INTERNSHIP PROGRAM

PROJECT NAME: STUDENT PERFORMANCE ANALYSIS & VISUALIZATION

TASK: TASK01- PYTHON PROGRAMMING,DATA ANALYSIS & VISUALIZATION

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INTERNSHIP DOMAIN: AI/ML

SUBMISSION DATE: 23-AUGUST-2026

1. Introduction:

This project is part of the Devixo Solutions AI/ML Internship Program – Task 01: Python Programming, Data Analysis & Visualization. The project focuses on analyzing student performance data using Python, data cleaning, exploratory data analysis, statistical analysis, and data visualization. The purpose of the project is to understand the dataset, identify important patterns and relationships, and generate meaningful insights from the students' academic performance data.

2. Objective:

The main objective of this project is to analyze student performance using Python and data analysis techniques. The project aims to clean and understand the dataset, perform exploratory and statistical analysis, create meaningful visualizations, identify important relationships and trends, and generate useful findings about student academic performance.

3. Dataset Name:

Student Performance Dataset

4. Dataset Source:

The dataset was obtained from Kaggle.

5. Dataset Description:

The Student Performance Dataset contains information about students' academic performance. It includes attendance, internal test scores, assignment scores, daily study hours, and final exam marks. The dataset is used to analyze student performance and identify important patterns and relationships among these variables.

6. Dataset Size:

The dataset contains 2,000 rows and 7 columns.

- Number of Rows: 2,000
- Number of Columns: 7

7. Features/Columns Description:

COLUMN	DESCRIPTION
Student_ID	Unique identification number of each student
Attendance_Percent	Student attendance percentage
Internal_Test_1	Score obtained in Internal Test 1
Internal_Test_2	Score obtained in Internal Test 2
Assignment_Score	Study assignment score
Daily_Study_Hours	Average daily study hours
Final_Exam_Marks	Marks obtained in the final examination

8. Methodology:

The project followed a complete data analysis workflow: data loading, data understanding, data cleaning, exploratory data analysis, statistical analysis, data visualization, insight generation, and final conclusion.

9. Data Loading:

The dataset was loaded into a Pandas DataFrame using Python in Jupyter Notebook. The first five records, last five records, and random sample of records were displayed to understand the dataset.

10. Data Cleaning:

The dataset was checked for missing values, duplicate records, incorrect data types, unclear column names, and invalid values. The data types and column names were verified and the cleaned dataset was checked again to ensure data quality before performing fur
ther analysis.

11.Exploratory Data Analysis:

Exploratory Data Analysis was performed to understand the distribution and characteristics of the dataset. Descriptive statistics such as mean, minimum, maximum, and standard deviation were examined. Relationships between important variables were also explored using different plots and statistical techniques.

12. Statistical Analysis:

Statistical analysis was performed on the numerical variables of the dataset. Measures such as mean, minimum, maximum, and standard deviation were calculated. Correlation analysis was also performed to understand the strength and direction of linear relationships between variables.

13. Data Visualizations:

Several visualizations were created to understand student performance and relationships between variables. The visualizations included histograms, bar charts, pie charts, scatter plots, line plots, pairplots, violin plots, box plots, and a correlation heatmap. These visualizations helped identify distributions, trends, relationships, and patterns in the dataset.

14. Code Screenshots

1.Required Libraries and Dataset Loading:

```
#1.1 Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#1.2 Load Data
data=pd.read_csv("Final_Marks_Data.csv")
```

This code imports the required Python libraries and loads the Student Performance dataset using Pandas.

2.Data Cleaning and Data Quality Checks:

```
#2.1 Check Missing Values
Missing_values=data.isnull().sum()
Missing_values
```

```
#2.3 Check Duplicate Records
data.duplicated().sum()
```

```
#2.4 Check Data TYpes
data.dtypes
```

```
#2.6 Check Invalid Values
data.min()
```

```
data.max()
```

```
#2.7 Verify the Cleaned Dataset
data.isnull().sum()
```

```
data.shape
```

```
data.describe()
```

This code checks the dataset for missing values, duplicate records, data types, and invalid values to ensure that the data is suitable for further analysis.

3.Exploratory Data Analysis:

```
#3.1 Descriptive Statistics
data.mean(numeric_only=True)
```

```
data.median(numeric_only=True)
```

```
data.mode(numeric_only=True)
```

```
data.var(numeric_only=True)
```

This code calculates summary statistics—including mean, median, mode, and variance—to measure central tendencies and data spread across numerical features.

4.Statistical Analysis Code:

```
data.min(numeric_only=True)
```

```
data.max(numeric_only=True)
```

```
data.std(numeric_only=True)
```

This code calculates minimum, maximum, and standard deviation values to determine data bounds and variability across student performance metrics.

5.Correlation Analysis Code:

```
correlation=data.select_dtypes(include=np.number).corr()
correlation
```

This code selects numerical columns and computes the correlation matrix to quantify linear relationships between academic features.

6.Visualization Code:

```
# Histogram
plt.figure(figsize=(8,5))
plt.hist(data["Final_Exam_Marks"],bins=10)
plt.title("Distribution of Final Exam Marks")
plt.xlabel("Final Exam Marks")
plt.ylabel("Number of Students")
plt.savefig("../images/visualization_01.png",bbox_inches="tight")
plt.show()
```

```
# Bar chart
avg_marks=data.groupby("Daily_Study_Hours")["Final_Exam_Marks"].mean()
plt.figure(figsize=(8,5))
plt.bar(avg_marks.index,avg_marks.values)
plt.title("Average Final Marks by Daily Study Hours")
plt.xlabel("Daily Study Hours ")
plt.ylabel("Average Final Marks")
plt.savefig("../images/visualization_02.png",bbox_inches="tight")
plt.show()
```

```
# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(data["Attendance_Percent"],
            data["Final_Exam_Marks"])
plt.title("Attendance vs Final Exam Marks")
plt.xlabel("Attendance Percentage")
plt.ylabel("Final Exam Marks")
plt.savefig("../images/visualization_04.png",bbox_inches="tight")
plt.show()
```

This code generates histograms, bar charts, and scatter plots using Matplotlib to visualize feature distributions, grouped averages, and variable relationships.

15:OUTPUT SCREENSHOTS

1:Data Cleaning Output

```
Student_ID                        0
Attendance (%)                    0
Internal Test 1 (out of 40)       0
Internal Test 2 (out of 40)       0
Assignment Score (out of 10)      0
Daily Study Hours                 0
Final Exam Marks (out of 100)     0
dtype: int64
```

```
np.int64(0)
```

```
Student_ID                        object
Attendance (%)                     int64
Internal Test 1 (out of 40)        int64
Internal Test 2 (out of 40)        int64
Assignment Score (out of 10)       int64
Daily Study Hours                  int64
Final Exam Marks (out of 100)      int64
dtype: object
```

```
Student_ID              S1000
Attendance_Percent         52
Internal_Test_1            18
Internal_Test_2            16
Assignment_Score            4
Daily_Study_Hours           1
Final_Exam_Marks           25
dtype: object
```

```
Student_ID              S2999
Attendance_Percent        100
Internal_Test_1            40
Internal_Test_2            40
Assignment_Score           10
Daily_Study_Hours           5
Final_Exam_Marks          100
dtype: object
```

```
Student_ID              0
Attendance_Percent      0
Internal_Test_1         0
Internal_Test_2         0
Assignment_Score        0
Daily_Study_Hours       0
Final_Exam_Marks        0
dtype: int64
```

```
(2000, 7)
```

	Attendance_Percent	Internal_Test_1	Internal_Test_2	Assignment_Score	Daily_Study_Hours	Final_Exam_Marks
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	84.891500	32.115500	32.464500	7.507000	2.823500	64.855000
std	7.758855	4.563504	4.522827	1.021015	0.608714	11.341265
min	52.000000	18.000000	16.000000	4.000000	1.000000	25.000000
25%	80.000000	29.000000	29.000000	7.000000	2.000000	58.000000
50%	85.000000	32.000000	33.000000	8.000000	3.000000	65.000000
75%	90.000000	35.000000	36.000000	8.000000	3.000000	73.000000
max	100.000000	40.000000	40.000000	10.000000	5.000000	100.000000

Explanation:

These output results confirm data integrity across all quality checks:

- **Completeness & Duplication:** Confirms 0 missing (null) values and 0 duplicate rows in the dataset.
- **Data Types & Range Validation:** Verifies correct data types (int64, object) for each feature and validates acceptable value bounds via minimum/maximum values.
- **Final Verification & Shape:** Displays the final dataset dimensions (shape) and generates basic summary statistics (describe()), ensuring the data is clean and ready for analytical modeling.

2. Exploratory Data Analysis:

```
Attendance_Percent    84.8915
Internal_Test_1       32.1155
Internal_Test_2       32.4645
Assignment_Score       7.5070
Daily_Study_Hours      2.8235
Final_Exam_Marks      64.8550
dtype: float64
```

```
Attendance_Percent    85.0
Internal_Test_1       32.0
Internal_Test_2       33.0
Assignment_Score       8.0
Daily_Study_Hours      3.0
Final_Exam_Marks      65.0
dtype: float64
```

	Attendance_Percent	Internal_Test_1	Internal_Test_2	Assignment_Score	Daily_Study_Hours	Final_Exam_Marks
0	86	32	35	8	3	66

```
Attendance_Percent     60.199828
Internal_Test_1        20.825573
Internal_Test_2        20.455968
Assignment_Score        1.042472
Daily_Study_Hours       0.370533
Final_Exam_Marks      128.624287
dtype: float64
```

Explanation:

These output results summarize the statistical distributions across student performance metrics:

- **Mean & Median:** Students average 84.89% attendance and 64.85 marks in final exams, with median values closely matching, indicating a balanced distribution without heavy skewness.
- **Mode & Variance:** Mode values reflect common performance benchmarks (e.g., 85% attendance, 65 exam marks), while variance shows how scores spread around the central average.

3. Statistical Analysis Code:

```
Attendance_Percent    52
Internal_Test_1       18
Internal_Test_2       16
Assignment_Score       4
Daily_Study_Hours      1
Final_Exam_Marks      25
dtype: int64
```

```
Attendance_Percent    100
Internal_Test_1        40
Internal_Test_2        40
Assignment_Score       10
Daily_Study_Hours       5
Final_Exam_Marks      100
dtype: int64
```

```
Attendance_Percent     7.758855
Internal_Test_1        4.563504
Internal_Test_2        4.522827
Assignment_Score       1.021015
Daily_Study_Hours      0.608714
Final_Exam_Marks      11.341265
dtype: float64
```

Explanation:

These output results detail the operational bounds and variability of student metrics:

- **Minimum & Maximum:** Establishes performance ranges across variables, showing attendance spans from 52% to 100%, study hours from 1 to 5 hours, and final exam marks from 25 to 100.
- **Standard Deviation:** Quantifies dispersion from average scores, indicating relatively low variation in daily study hours (std = 0.61) and higher spread in final exam marks (std = 11.34).

4. Correlation Analysis Output:

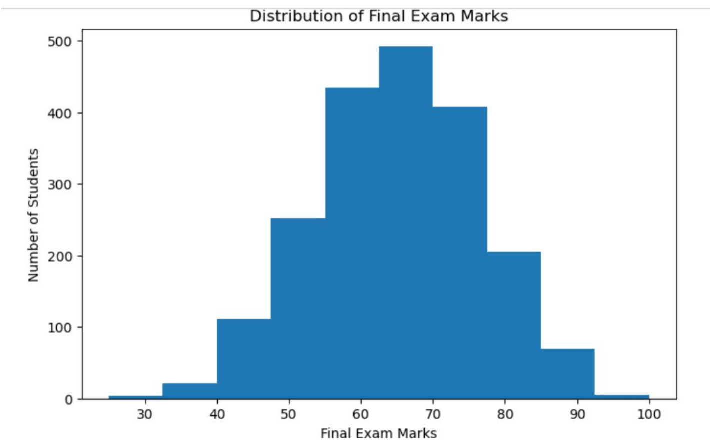
	Attendance_Percent	Internal_Test_1	Internal_Test_2	Assignment_Score	Daily_Study_Hours	Final_Exam_Marks
Attendance_Percent	1.000000	0.505033	0.494560	0.432373	0.298767	0.725644
Internal_Test_1	0.505033	1.000000	0.263400	0.568047	0.213539	0.689227
Internal_Test_2	0.494560	0.263400	1.000000	0.541970	0.225670	0.691049
Assignment_Score	0.432373	0.568047	0.541970	1.000000	0.169005	0.669400
Daily_Study_Hours	0.298767	0.213539	0.225670	0.169005	1.000000	0.412877
Final_Exam_Marks	0.725644	0.689227	0.691049	0.669400	0.412877	1.000000

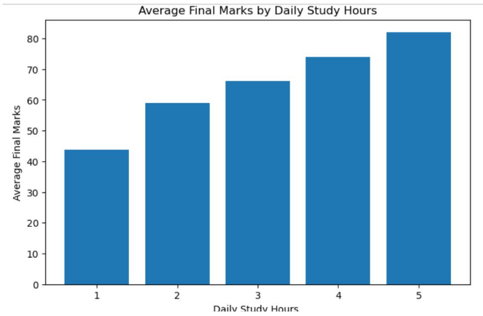
Explanation:

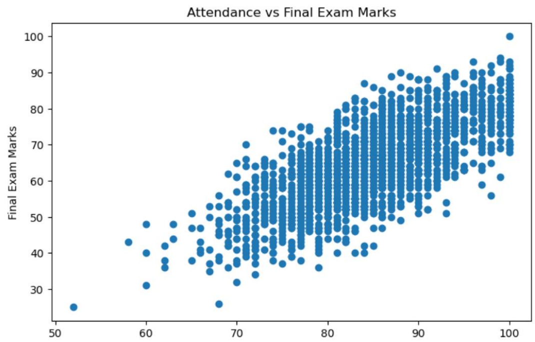
These output results detail the operational bounds and variability of student metrics:

- **Minimum & Maximum:** Establishes performance ranges across variables, showing attendance spans from 52% to 100%, study hours from 1 to 5 hours, and final exam marks from 25 to 100.
- **Standard Deviation:** Quantifies dispersion from average scores, indicating relatively low variation in daily study hours (std = 0.61) and higher spread in final exam marks (std = 11.34).

5. Visualization Code:







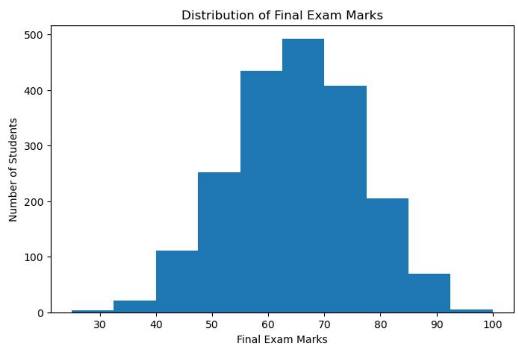
Explanation:

These visualizations reveal clear distributional patterns and positive academic trends across the dataset:

- **Histogram (Exam Marks Distribution):** Displays a bell-shaped, normal distribution centered around the 60–70 mark range, indicating balanced student performance.
- **Bar Chart (Marks by Study Hours):** Illustrates a steady upward progression, showing average exam marks rising directly from ~44 marks at 1 study hour to over 80 marks at 5 study hours.
- **Scatter Plot (Attendance vs Marks):** Demonstrates a strong positive linear relationship, confirming that higher attendance percentages consistently correspond to higher final exam scores.

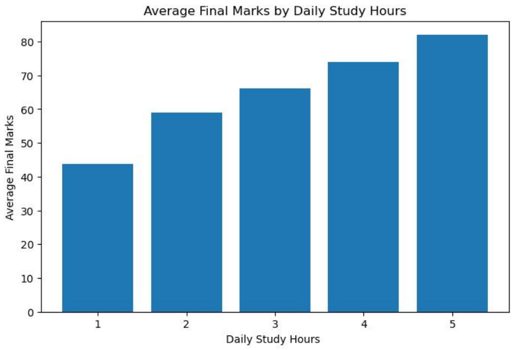
16.GRAPH SCREENSHOTS

Figure 1: Distribution of Final Exam



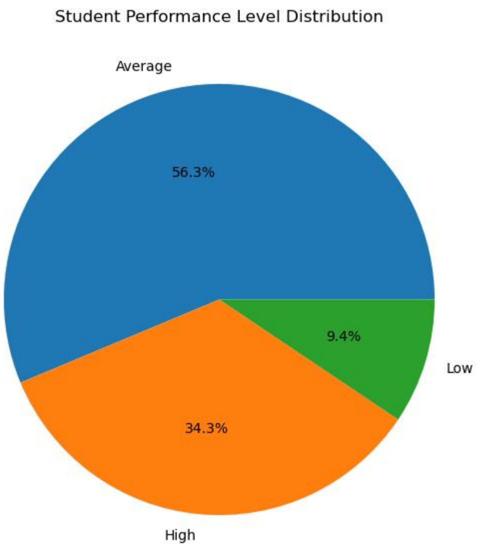
Interpretation:
The histogram shows that most students scored between 55 and 80 marks. The highest concentration of students is around 65–70 marks.

Figure2:Average Final Marks by Daily Hours



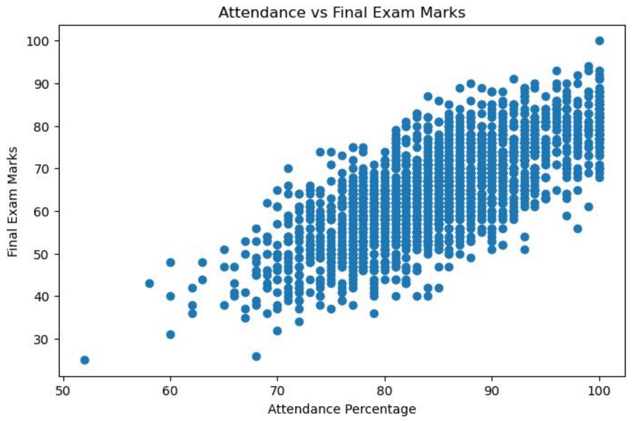
Interpretation:
The bar chart shows that average final exam marks increase as daily study hours increase. Students who study more generally achieve higher average marks.

Figure3: Student Performance Level Distribution



Interpretation:
The pie chart shows that most students are in the Average performance category (56.3%). The High category represents 34.3% of students, while the Low category represents 9.4%.

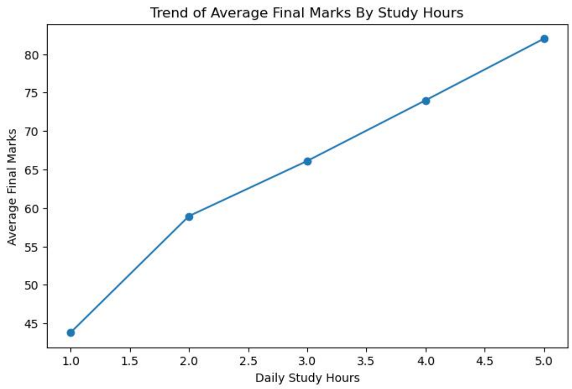
Figure4:Attendance vs Final Exam Marks



Interpretation:

The scatter plot shows a positive relationship between attendance and final exam marks. Students with higher attendance generally tend to achieve higher marks.

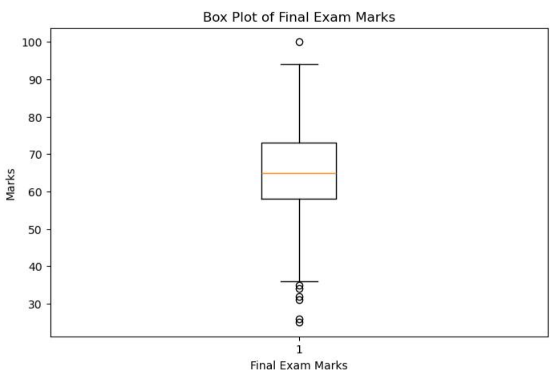
Figure5:Trend of Average Final Marks By Study Hour



Interpretation:

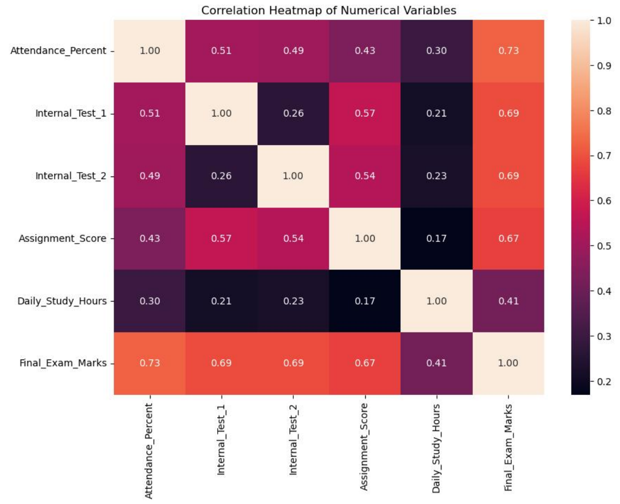
The line chart shows an increasing trend in average final exam marks as daily study hours increase. This indicates that higher study time is associated with better academic performance.

Figure 6:Box Plot of Final Exam Marks



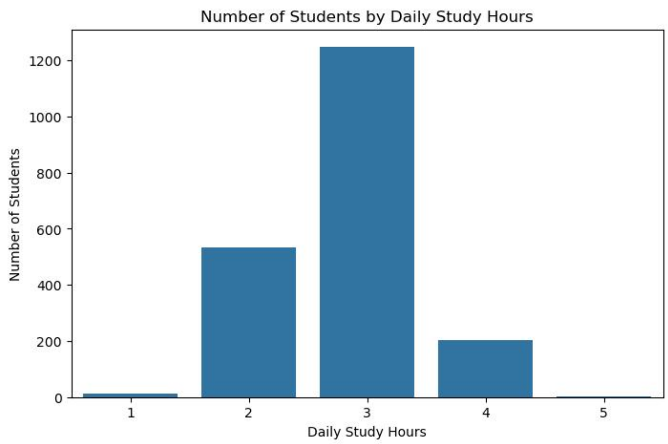
Interpretation:
The box plot shows a median of approximately 65 marks and an interquartile range from about 55 to 73. Some possible outliers are visible at the lower and higher ends.

Figure 7:Correlation Heatmap of Numerical Variables



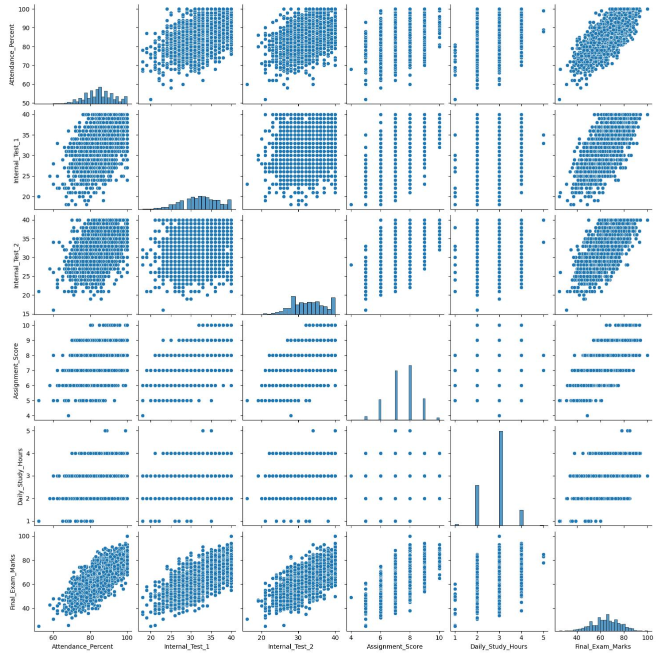
Interpretation:
Attendance (0.73) and internal tests (0.69) show the strongest relationships with Final Exam Marks. Daily Study Hours (0.41) show a weaker relationship with final exam marks.

Figure 8:Number of Students by Daily Study Hours



Interpretation:
Most students study 3 hours daily, followed by 2 hours. Very few students study for 1 hour or 4 or more hours per day.

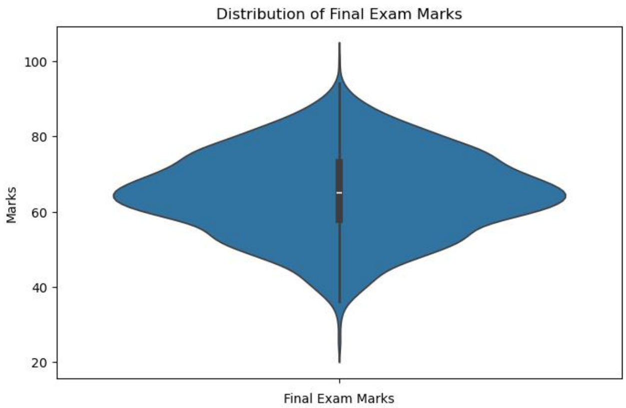
Figure 9: Pair Plot



Interpretation:

The pair plot shows positive relationships among attendance, internal test scores,assignment scores, daily study hours,and final exam marks. The upward trends indicate that higher values of these variables are generally associated with higher final exam marks. The diagonal plots show the distribution of each variable.

Figure 10: Distribution of Final Exam Marks



Interpretation:

The violin plot shows the distribution of Final Exam Marks.The distribution is centered around a median of approximately 65, with most students concentrated around the average score range.

17. INSIGHTS & MAJOR FINDINGS:

1. **Average Final Exam Marks:** The average is 64.86, with a median of 65.
2. **Final Exam Range:** Marks range from 25 to 100.
3. **Average Attendance:** Average attendance is 84.89%, ranging from 52% to 100%.
4. **Internal Tests:** Average scores are 32.12/40 in Test 1 and 32.46/40 in Test 2.
5. **Assignment Score:** The average assignment score is 7.51/10.
6. **Daily Study Hours:** Students study 2.82 hours per day on average.
7. **Study Hours Trend:** More study hours are associated with higher average final marks.
8. **Attendance Trend:** Higher attendance is generally associated with higher final exam marks.
9. **Performance Distribution:** 56.3% of students are in the Average category, 34.3% in the High category, and 9.4% in the Low category.
10. **Most Common Mark:** The most frequently occurring Final Exam Mark is 66.
11. **Mark Distribution:** Most final marks are concentrated around 55–80.
12. **Overall Finding:** Regular attendance and more study time are positively associated with better performance.

18. CONCLUSION:

This project analyzed a dataset of 2,000 student records to understand student academic performance. The dataset included attendance, internal test scores, assignment scores, daily study hours, and final exam marks. The data was cleaned by checking missing values, duplicate records, data types, column names, and invalid values, followed by verification of the cleaned dataset. Exploratory and statistical analysis was performed using mean, median, mode, minimum, maximum, range, standard deviation, variance, and correlation analysis. Different visualizations, including histograms, bar charts, pie charts, scatter plots, line charts, correlation heatmaps, pairplots, violin plots, and box plots, were created to identify patterns and relationships in the data.

The analysis showed that attendance and internal test scores had strong positive relationships with final exam marks, while daily study hours also showed a positive relationship with performance. Most students were in the Average performance category, and final exam marks were mainly concentrated around the middle range. Through this project, I learned how to load, clean, analyze, visualize, and interpret real-world student data using Python and its data analysis libraries. This type of analysis can be useful in educational settings to understand student performance, identify important factors affecting results, and support better academic planning and decision-making.